

Q.1. Prove that $c^* = \left(\frac{2k_B T}{m} \right)^{1/2}$

Soln: We start with MB-distribution formula -

$$f(c) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}}$$

The most probable speed is at the peak or maximum of the speed distribution curve.

i.e., $\frac{df}{dv} = 0$ at the maximum.

to simplify it mathematically, take $\left\{ \begin{array}{l} A = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \\ B = \frac{m}{2k_B T} \end{array} \right.$

i.e., $f(c) = A v^2 e^{-Bv^2}$

Applying product rule,

$$\frac{df}{dv} = A \left[2v \cdot e^{-Bv^2} + v^2 (-2Bv) e^{-Bv^2} \right] = 0$$

i.e., $A \cdot e^{-Bv^2} \cdot 2v (1 - Bv^2) = 0$

for non-zero speed $v \neq 0$; therefore, we can set the term in parenthesis to zero -

$$1 - Bv^2 = 0$$

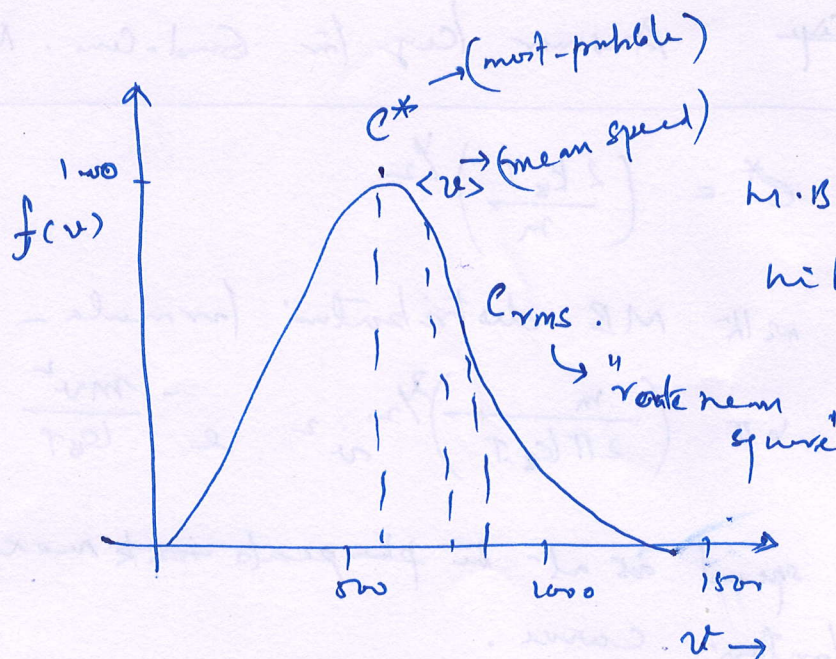
i.e., $v^2 = \frac{1}{B} = \frac{1}{m/2k_B T} = \left(\frac{2k_B T}{m} \right)$

$\therefore v = \left(\frac{2k_B T}{m} \right)^{1/2}$

(3 Marks)

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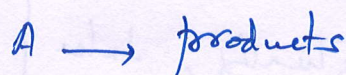


M.B. distribution for nitrogen molecules.

(2 Marks)

Q.2.

for zero-order rxn



$$\frac{dx}{dt} = k$$

Upon integration

$$\int dx = k \int dt$$

i.e., $x = kt + \text{const.}$

2 Marks

i.e., $x = kt$ and $\frac{x}{t} = k$ | x is zero for $x=0$ at $t=0$ ✓

in the question $x = C_A^0/2$ for 50%.

and $k = \frac{C_A^0}{2 t_{1/2}} = \frac{C_A^0}{2 \times 20}$

By, $x = \frac{3 C_A^0}{4}$ for 75% completion.

3 Marks

i.e., $k = \frac{3/4 C_A^0}{t_{1/2}}$

or, $\frac{C_A^0}{40} = \frac{3/4 C_A^0}{t}$

i.e., $t = \frac{40 \times 3}{4}$

= 30 minutes

Q. 3.

M-M. eqn for Enzymatic rethⁿ in

$$\frac{1}{v_0} = \frac{1}{v_m} + \frac{K_m}{v_m} \frac{1}{S_0} \quad \checkmark$$

$$v_m = k_2 E_0 \quad \times$$

$$K_m = \frac{k_{-1} + k_2}{k_1}$$

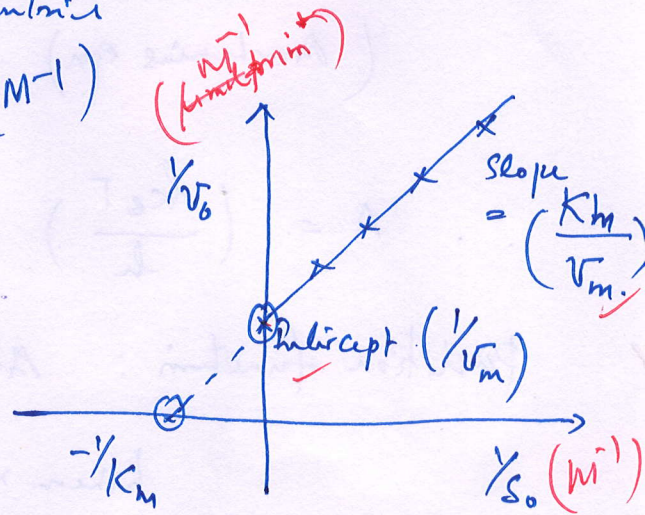
Now, Y-axis entails

$$\left(\frac{1}{v_0} \right) \left(\frac{M}{\cancel{\text{mol}} \cancel{\text{L}}^{-1} \text{min}^{-1}} \right)$$

X-axis entails

$$\left(\frac{1}{S_0} \right) (M^{-1})$$

2.8	_____	0.7
3.2	_____	0.9
4.2	_____	1.3
6.9	_____	2.2
9.0	_____	3.3



(Graph - 2 Marks)

1 mark
Report

$$\text{slope} = \left(\frac{K_m}{v_m} \right) =$$

$$0.12625 \text{ min}^{-1}$$

$$K_m = 0.116 \text{ M}$$

1 mark
Intercept

$$= \frac{1}{v_m}$$

$$1.0885$$

$$v_m = 0.9187 \text{ M min}^{-1}$$

2 marks
k₂ (TOF)

$$= \frac{v_m}{E_0} =$$

$$\frac{0.9187 \text{ min}^{-1}}{E_0}$$

Report all
in units
if unit is
not mentioned
because 1/2
mark each.

Q.4.

Atom + Linear molecule (diatomic) $\rightarrow$ Non-linear molecule.

We know,

$$k = A e^{-E_a/RT} = \left(\frac{k_B T}{h} \right) \frac{Q^\ddagger}{Q_A Q_B} e^{-E_a/RT}$$

(Arrhenius eqn) (TST in terms of partition function)

$$\therefore A = \left(\frac{k_B T}{h} \right) \frac{Q^\ddagger}{Q_A Q_B}$$

Partition function: Atom $Q_A = q_t^3$

Linear molecule $Q_B = q_t^3 q_r^2 q_v^1$

3 Marks
till this part

$$Q^\ddagger (A \cdots B) \text{ (non linear triatomic)} = q_t^3 q_r^3 q_v^{(3-1)} = q_t^3 q_r^3 q_v^2$$

$$A = \frac{1.38 \times 10^{-23} \text{ (J/K)} \times 298 \text{ K}}{6.626 \times 10^{-34} \text{ (J s)}} \times \frac{(10^8)^3 (10)^3 (1)^2}{(10^8)^3 (10^8)^3 (10)^2 (1)}$$

2 marks
till reporting A = $62.12 \times 10^{-8} \text{ s}^{-1}$ ✓
like units
Now convert in to $\text{M}^{-1} \text{s}^{-1}$ ✓

$$= 62.12 \times 10^{-8} \times 6.023 \times 10^{23} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

$$= 3.79 \times 10^{10} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

$$= 3.79 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$$

Q.5.

Oxygen molecule at 600K.

$$O_2^{(mol)} = 32 \text{ amu}$$

$$O_2^{(diam)} = 346 \text{ pm.}$$

Soln: Start with $\langle v \rangle = \left(\frac{8k_B T}{\pi \mu} \right)^{1/2}$

$$= \left[\frac{8 \times 1.38 \times 10^{-23} \text{ (J/K)} \times 600 \text{ K} \times 6.023 \times 10^{23} \text{ (amu/g)} \times 1000 \text{ (g/kg)}}{3.14 \times 32 \text{ amu.}} \right]^{1/2}$$

$$= [3,97,057]^{1/2}$$

$$= \frac{630.2 \text{ m/s}}{\quad} \quad \frac{630.16 \text{ m/s}}{\quad}$$

$$= \underline{630.125 \text{ m/s}}$$

$$C_v = \frac{5R}{2} = \frac{5 \times 8.314 \text{ (J/mol/K)}}{2}$$

$$= \underline{20.8 \text{ J/mol/K}}$$

$$k = \frac{\langle v \rangle C_v}{3\sqrt{2} \pi r_{max}^2 N_A}$$

$$= \frac{630.125 \text{ (m/s)} \times 20.8 \text{ (J/mol/K)}}{3 \times \sqrt{2} \times 3.14 \times (346 \times 10^{-12} \text{ m})^2 \times 6.023 \times 10^{23} \text{ molecules/mol}}$$

$$= 1.36 \times 10^{-2} \text{ J s}^{-1} \text{ m}^{-1} \text{ K}^{-1}$$

$$= \underline{1.36 \times 10^{-2} \text{ W m}^{-1} \text{ K}^{-1}}$$

Q.6.

$$J_z = -D \left(\frac{\partial n^*}{\partial z} \right)$$

and $pV = nRT$; $n = \frac{pV}{RT}$

here, $\left(\frac{\partial n^*}{\partial z} \right) = \frac{80 (\text{Torr/m}) \times (1 \text{ atm} / 760 \text{ Torr}) \times 6.023 \times 10^{23} (\text{molecules/mole}) \times 1000 \text{ L/m}^3}{0.082 (\text{L atm mol}^{-1} \text{K}^{-1}) \times 273 \text{ K}}$

2 marks

$$= 2.07 \times 10^{24} \text{ molecules/m}^4$$

$$\therefore J_z = -D \left(\frac{\partial n^*}{\partial z} \right) = 1.8 \times 10^{-5} (\text{m}^2 \text{ s}^{-1}) \times 2.07 \times 10^{24} \text{ molecules/m}^4$$

1 mark

$$= 3.73 \times 10^{19} \text{ molecules sec}^{-1} \text{ m}^{-2}$$

finally, number of molecules, crossing area 0.4 m^2

$$|J_z| A = (0.4 \text{ m}^2) \times 3.73 \times 10^{19} \text{ molecules sec}^{-1} \text{ m}^{-2}$$

$$= 1.49 \times 10^{+19} \text{ molecules/second.}$$

2 marks

Q.7. (a)

$$D = \frac{k_B T}{6\pi \eta a}$$

$$T = 298 \text{ K}$$

$$a = 1 \text{ nm} = 1 \times 10^{-9} \text{ m}$$

$$\eta = 0.8 \text{ cPa} = 0.8 \times 0.001 \text{ Pa}\cdot\text{sec}$$

$$\begin{aligned} \therefore D &= \frac{1.38 \times 10^{-23} (\text{J/K}) \times (298 \text{ K})}{6 \times 3.14 \times (0.8 \times 0.001 \text{ N/m}^2\cdot\text{sec}) \times (1 \times 10^{-9} \text{ m})} \\ &= \underline{\underline{2.73 \times 10^{-10} \text{ m}^2 \text{ sec}^{-1}}} \end{aligned}$$

(b) $k_{\text{enc}} = 4\pi (D_A + D_B) R = 4\pi (2D_A) R$

$$D_A = 2.73 \times 10^{-10} \text{ m}^2 \text{ sec}^{-1}$$

$$R = 2 \times 10^{-9} \text{ m}$$

$$\begin{aligned} \therefore k &= 4 \times 3.14 \times 2 \times (2.73 \times 10^{-10} \text{ m}^2 \text{ sec}^{-1}) \times \\ &\quad (2 \times 10^{-9} \text{ m}) \times 6.023 \times 10^{+23} \frac{\text{molecules}}{\text{mole}} \\ &\quad \times \left(10^{+6} \frac{\text{cm}^3}{\text{m}^3} \right) \end{aligned}$$

2.5 Marks

$$= 826 \times 10^{10} \text{ cm}^3 \text{ mol}^{-1} \text{ sec}^{-1}$$

$$= \underline{\underline{0.826 \text{ M}^{-1} \text{ sec}^{-1} \times 10^{10} \text{ M}^{-1} \text{ cm}^3}}$$

$$\begin{array}{r} -10 + 23 - 9 + 6 \\ \hline 29 - 19 \\ \hline = 10^{10} \\ \hline \text{M}^{-1} = \text{dm}^3 \text{ mol}^{-1} \end{array}$$

Q.8.

Schematically :-

Marcus Inverted region.

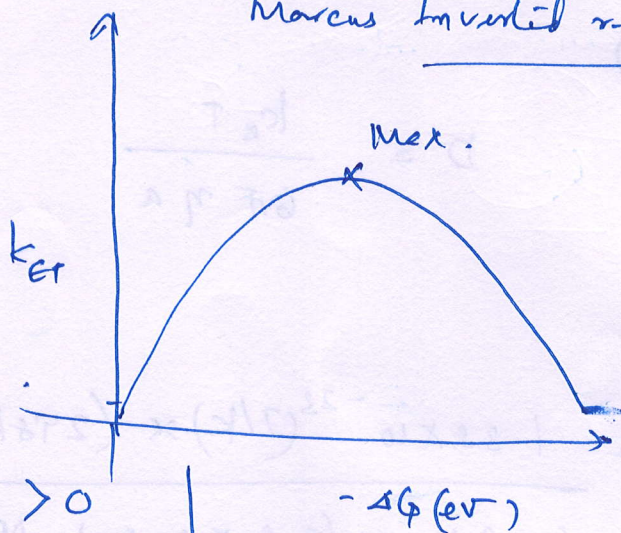
(a)

On increasing $|E - \Delta G|$

ET rate increases, and peaks.

Further drops.

$$\left| \begin{array}{l} \text{i) } \Delta G^\circ \\ E_m + \Delta G^\circ > 0 \\ \text{ii) } \Delta G^\circ \\ E_m + \Delta G^\circ < 0 \end{array} \right|$$



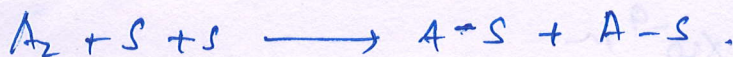
(2 Marks)

(b) prove mathematically

$$k_{ET}(T) = \left(\frac{k_B T}{h} \right) \exp \left\{ - \frac{(E_m + \Delta G^\circ)^2}{4 E_m k_B T} \right\}$$

(3 Marks)

Q.9



rate of absorption $r_a = k_a A_2 (1-\theta)^2$

rate of desorption $r_d = k_d \theta^2$

at eqm.

$$k_a A_2 (1-\theta)^2 = k_d \theta^2 \quad \left| \quad K = \frac{k_a}{k_d} \right.$$

$$\therefore K A_2 = \frac{\theta^2}{(1-\theta)^2}$$

$$\sqrt{K A_2} = \frac{\theta}{1-\theta}$$

$$\theta = \sqrt{K A_2} (1-\theta)$$

$$\theta = \sqrt{K A_2} - \sqrt{K A_2} \theta$$

$$\therefore \theta (1 + \sqrt{K A_2}) = \sqrt{K A_2}$$

$$\therefore \theta = \frac{\sqrt{K A_2}}{1 + \sqrt{K A_2}}$$

$$\therefore \theta \propto A_2^{1/2}$$

Q.10

(a)

3 Marks

$$\frac{\phi_f^\circ}{\phi_f} = 1 + K_{sr} \phi$$

Derive:

(1 + 2 Marks)

$$K_{sr} = k_q \cdot \tau$$

i, the rate of quenching goes proportionately with the number of quenchers, life time of the process and the quenching rate of the reaction.

(b) $K_{sr} = 10^2 \text{ M}^{-1}$

$\tau = 10 \text{ ns} = 10 \times 10^{-9} \text{ sec.}$

2 Marks $\therefore k_q = \frac{10^2 \text{ M}^{-1}}{10 \times 10^{-9} \text{ s}} = \underline{\underline{10^{10} \text{ M}^{-1} \text{ sec}^{-1}}}$